

US geneticist and zoologist Thomas Hunt Morgan won a Nobel Prize for his groundbreaking work on heredity. Following the work of Gregor Mendel, he confirmed the role of chromosomes and laid the foundations for the modern field of genetics.

#### MILESTONES

##### EARLY EXPERIMENTS

Accepts teaching post at Bryn Mawr in 1891; works in experimental embryology focusing on differentiation.

##### THE FLY ROOM

In 1904, moves to Columbia University; begins breeding program for fruit fly experiments in 1908.

##### SEMINAL TEXT

Publishes landmark text describing how heredity functions according to Mendel's laws in 1915.

##### MARINE RESEARCH

Becomes head of biology at Caltech in 1928; helps establish marine biology research unit.

##### PRIZE WINNER

Receives Nobel Prize in Physiology or Medicine for his work on inherited characteristics in 1933.

**Morgan bred** millions of fruit flies in his "Fly Room" laboratory. Able to produce a set of offspring in around 10 days, fruit flies were ideal for studying inheritance.

Thomas Hunt Morgan was born to a wealthy family in Kentucky in 1866. His countryside upbringing fueled a lifelong fascination with the natural world, from collecting fossils as a boy to conducting biological fieldwork during adolescence. Aged just 16, Morgan attended the University of Kentucky to study sciences before accepting a place to carry out postgraduate work in morphology and physiology at Johns Hopkins University. By 1890, aged 24, he had gained his PhD in zoology. In 1891, he became associate professor of biology at Bryn Mawr College, where he combined teaching with research into embryology. During his tenure at Bryn Mawr, Morgan made significant advances in experimental embryology, although it was to be his later work in genetics that earned him lasting fame.

#### The question of inheritance

In 1904, Morgan moved to New York to take up the role of professor of experimental zoology at Columbia University, and it was here that his groundbreaking work in heredity began. At the time, the field of genetics, as it is now known, did not exist—research had been conducted into inherited characteristics, but much of this had been inconclusive or partial. In 1866, the Augustinian friar Gregor Mendel (see pp.140–143) had theorized about "particles of inheritance"—tiny particles that passed traits down between generations—after studying inherited characteristics in pea plants.

